

01.119/99.503: Nanoelectronics and Technology

Homework 1 –Brief Solutions

1. Consider the 1R + 1 n-MOSFET circuit as shown in Fig. 1.
 - (a) Draw the comparative DC load lines for $R_D=5\text{ k}\Omega$ and $R_D=25\text{ k}\Omega$ on the I_D - V_{DS} characteristics (Fig. 2). Assume $V_{DD}=2.5\text{V}$.
 - (b) Draw the V_{in} Vs V_o (Transfer) characteristics for $R_D=5\text{ k}\Omega$ and $R_D=25\text{ k}\Omega$ and explain.
 - (c) Explain why 1R+ 1 n-MOSFET circuit will not work for real applications on Silicon technology?

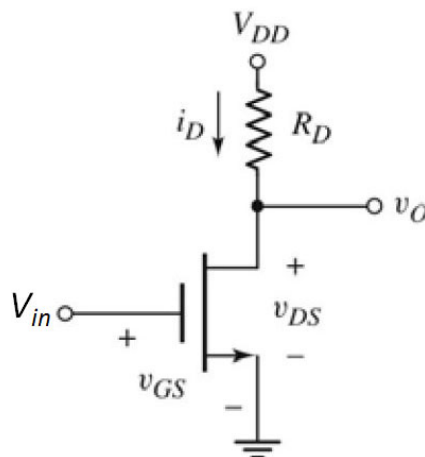


Fig. 1

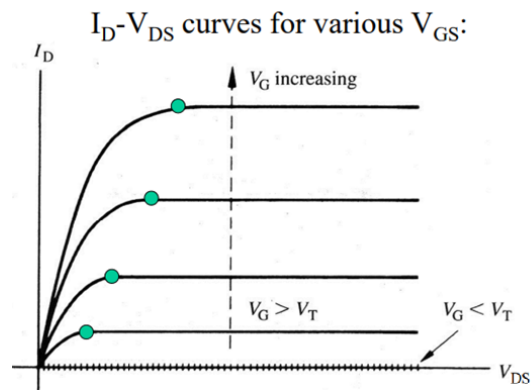
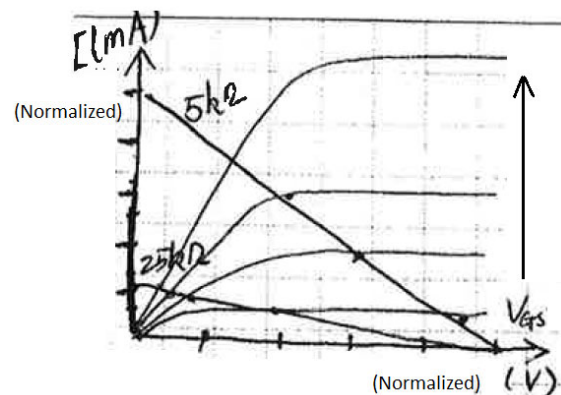


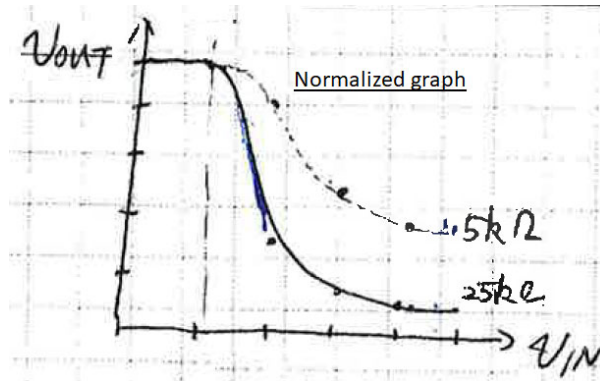
Fig. 2

Answer:

(a)



(b)



(c)

- Fabrication issue on Silicon substrate
- Decrease in drive current
- Power loss (I^2R)
- Increased static power

2. (a) What is meant by **skewing** in the CMOS inverter transfer characteristic?

Ans:

A skewed CMOS inverter has an **asymmetrical voltage transfer characteristic (VTC)** caused by an imbalance between the driving strengths of the PMOS and NMOS transistors. This shifts the switching threshold (V_{inv}) away from the ideal center point of $V_{DD}/2$.

Causes of skewing

Skewing is determined by the β -ratio, which is the ratio of the PMOS gain factor (β_p) to the NMOS gain factor (β_n). ϕ

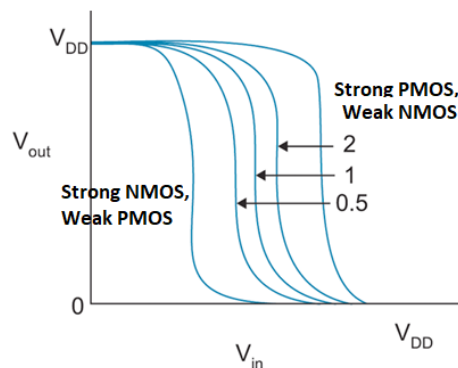
- $\beta_p = \mu_p C_{ox} (W/L)_p$
- $\beta_n = \mu_n C_{ox} (W/L)_n$

Where μ is the carrier mobility, C_{ox} is the gate oxide capacitance, and W/L is the transistor width-to-length ratio. Skewing occurs when the β -ratio is not equal to 1.

- **Unequal transistor sizing (W/L):** The mobility of electrons in an NMOS transistor (μ_n) is roughly 2 to 3 times greater than the mobility of holes in a PMOS transistor (μ_p). To achieve a symmetric VTC with a switching threshold at $V_{DD}/2$, the PMOS transistor is typically made wider than the NMOS transistor to match their current-driving capabilities. A deliberate mismatch in these sizes is the most common way to create a skewed inverter.
- **Process variations:** During manufacturing, variations in parameters like channel length, width, and threshold voltage can cause mismatches in the transistor characteristics, leading to an unintended skewed VTC.
- **Aging effects:** Stress from prolonged operation can degrade transistor performance over time, causing a shift in the threshold voltage and creating a skewed characteristic.
- **High-skew inverter ($\beta_p/\beta_n > 1$):**
 - This occurs when the PMOS transistor is made stronger (more dominant) than the NMOS transistor.
 - The VTC is shifted to the **right** (towards V_{DD}), meaning the switching threshold (V_{inv}) is greater than $V_{DD}/2$.
- **Low-skew inverter ($\beta_p/\beta_n < 1$):**
 - This occurs when the NMOS transistor is made stronger (more dominant) than the PMOS transistor.
 - The VTC is shifted to the **left** (towards ground), meaning the switching threshold (V_{inv}) is less than $V_{DD}/2$.

skewed gate

- If $\beta_p / \beta_n \neq 1$, switching point will move from $V_{DD}/2$



(b) Why does a skewed inverter improve **noise margin** for one logic level but degrade it for the other? You consider the case of a **strong NMOS and weak PMOS**.

Ans:

When NMOS is Stronger than PMOS

When the NMOS transistor is stronger (larger W_n/L_n or higher mobility), it can **pull the output low more effectively** than the PMOS can pull it high.

That means:

- The inverter switches from HIGH $\rightarrow$ LOW at a **lower input voltage**.
- The **switching threshold V_M shifts to the left** (below $V_{DD}/2$).

Effect on Noise Margins

Let's recall:

$$NML = V_{IL} - V_{OL}, \quad NMH = V_{OH} - V_{IH}$$

- $V_{OL} \approx 0$
- $V_{OH} \approx V_{DD}$

Now, when V_M moves **left**:

(a) Noise Margin for Logic '0' (NML) Increases

- The **low-input region** (where $V_{in} < V_{IL}$) becomes **wider**.
- A larger range of small input voltages still produces a full HIGH output.
- Hence, **logic '0' becomes more tolerant to noise**.

$\rightarrow$ **NML increases.**

(b) Noise Margin for Logic '1' (NMH) Decreases

- The **high-input region** (where $V_{in} > V_{IH}$) becomes **narrower**.
- A small downward fluctuation in input voltage (noise) could wrongly drive the output HIGH again.
- Hence, **logic '1' becomes less tolerant to noise**.

$\rightarrow$ **NMH decreases.**

Physical Interpretation

A **strong NMOS** quickly discharges the output (low resistance path to ground), but the **weak PMOS** struggles to charge the output fully.

Thus, the circuit becomes:

- More robust in recognizing logic '0's (strong pull-down),
- More sensitive when recognizing logic '1's (weak pull-up).

3. (a) Explain why **short-circuit current** is higher for **slow input rise/fall times** compared to fast switching.

(b) How does **load capacitance** influence dynamic power consumption?

Ans:

(a)

In a CMOS inverter, during switching, there is a brief interval when **both NMOS and PMOS conduct simultaneously**.

- The PMOS tries to pull the output up to V_{DD} .
- The NMOS tries to pull it down to 0 V.

During this overlap, a **direct current path** from $V_{DD} \rightarrow \text{PMOS} \rightarrow \text{NMOS} \rightarrow \text{GND}$ is formed.

That transient current is called **short-circuit current**

Slow input transition

- The input voltage sweeps gradually from 0 V $\rightarrow V_{DD}$.
- Around V_M , both transistors stay partially ON for a **longer time**.
- A significant current flows directly from V_{DD} to GND through both devices.

Short-circuit current increases

because **the overlap conduction time increases**.

Fast input transition

- The input quickly crosses the switching threshold ($V_M \approx V_{DD}/2$).
- The overlap of conduction between PMOS and NMOS is **very short**.
- The output changes rapidly $\rightarrow$ less energy lost through short-circuit path.

Short-circuit current (and power) is small.

(b)

Dynamic power is the power consumed when a CMOS circuit **charges and discharges capacitive loads** (mainly gate and interconnect capacitances).

$$P_{dynamic} = \alpha C_L V_{DD}^2 f$$

Role of Load Capacitance (C_L)

Every CMOS inverter or logic gate drives some **capacitance** at its output:

When the output switches:

- The PMOS **charges** C_L from $0 \rightarrow V_{DD}$
- The NMOS **discharges** C_L from $V_{DD} \rightarrow 0$

Each transition involves movement of charge $Q = C_L V_{DD}$.

Energy per Transition

Energy used to charge the capacitor:

$$E = \frac{1}{2} C_L V_{DD}^2$$

Since every complete cycle (charge + discharge) dissipates this energy in the pull-up and pull-down networks, the **average power** is:

$$P = \alpha f C_L V_{DD}^2$$

Thus, **dynamic power increases linearly with C_L** .

4. Explain the impact of constant electric field scaling of CMOS on threshold voltage (V_{th}).

Ans:

Constant-field scaling keeps electric field constant $\rightarrow V_{DD}$ decreases by $1/\kappa$.

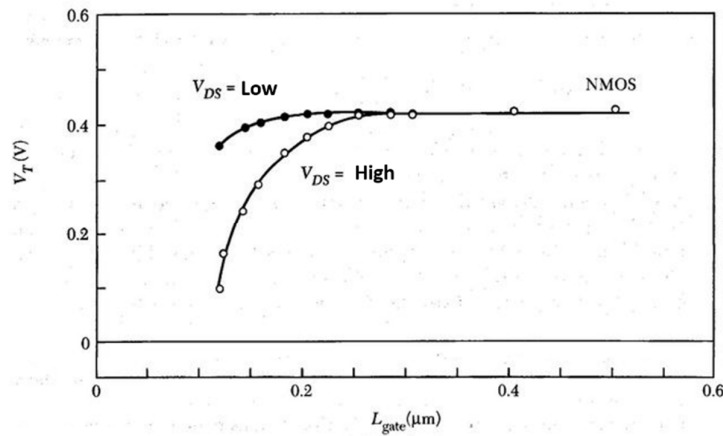
If V_{th} were not reduced, the ratio V_{DD}/V_{th} would drop and the transistor might not switch properly.

Therefore, V_{th} must also **reduce slightly**, though not linearly, to maintain:

- **Proper logic swing,**
- **Adequate drive current, and**
- **Acceptable subthreshold leakage.**

However, too small $V_{th} \rightarrow$ high **off-state leakage**, so practical designs often deviate from ideal constant-field scaling.

With gate length scaling, barrier lowering (BL) resulting in threshold voltage roll-off with channel length reduction. Drain induced barrier lowering (DIBL) reduces threshold voltage for short channel devices and increases threshold voltage roll-off.



5. List the key challenges of CMOS scaling from Micro to Nano CMOS (till 32nm Technology). Briefly explain the solutions for the respective challenges.

Answer:

- Threshold voltage variation
- DIBL (Drain Induced Barrier Lowering)
- GIDL (Gate Induced Drain Leakage)
- Shallow S/D
- Mobility issues/Velocity saturation/Hot carrier effect
- Gate oxide thinning/leakage current
- High-K/Metal Gate-Reliability issues

6. With the help of schematics (Cross-sectional view) of Long-Channel MOSFET (Micrometer technology) and Short-channel MOSFET, highlight the changes in the structure, material, doping and other parameters in these two structures.

SCALING OF MOSFET

